METRIC

MIL-PRF-32167

29 July 2004

PERFORMANCE SPECIFICATION

TRANSIENT VOLTAGE SURGE SUPPRESSOR (TVSS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the requirements for a transient voltage surge suppressor (TVSS) device used to provide protection from electrical system transients for electrical equipment installed aboard U.S. Navy ships.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/26307	-	Gray, Semi-Gloss

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-E-917	-	Electric Power Equipment Basic Requirements
MIL-DTL-15090	-	Enamel, Equipment, Light Gray, (Navy Formula No. 111)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subjects line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 5920

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461		Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1310	-	Standard Practice for Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety
MIL-STD-1399-300	-	Interface Standard for Shipboard Systems, Section 300A, Electric Power, Alternating Current (Metric)
DOD-STD-1399-301	-	Interface Standard for Shipboard Systems, Section 301A, Ship Motion and Attitude (Metric)

(Copies of these documents are available online at http://astimage.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NSTM Chapter 631	-	Preservation of Ships in Service

(Copies of this document are available online at http://fas.org/man/dod-101/navy/docs/index.html or from the Federation of American Scientists 1717 K. Street, NW, Suite 209, Washington DC 20036.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ISO/ASQ Q9001	-	Quality Management Systems - Requirements

(Copies of this document are available online at www.asq.org or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53201-1734.)

ASTM INTERNATIONAL

F1507	-	Standard Specification for Surge Suppressors for Shipboard Use

(Copies of this document are available online at www.astm.org or from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 45	-	IEEE Recommended Practice for Electric Installations on Shipboard
IEEE C62.41.1	-	IEEE Guide on the Surge Environment in Low-Voltage (1000 V and less) AC Power Circuits
IEEE C62.41.2	-	IEEE Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and less) AC Power Circuits

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IEEE C62.45	-	IEEE Recommended Practice on Surge Testing for Equipment Connected to Low-Voltage (1000V and Less) AC Power Circuits

(Copies of these documents are available online at www.ieee.org or from the Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA 250	-	Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA LS-1	-	Low Voltage Surge Protective Devices

(Copies of this document are available online at www.nema.org or from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 94	-	Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 1449	-	Transient Voltage Surge Suppressors

(Copies of these documents are available online at www.ul.comor from the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. A TVSS device is an electronic device that is added to an electrical circuit to provide protection to the load from transient voltages in the system. This is accomplished by attenuating or reducing, to an acceptable level, the magnitude of electrical distribution system disturbances such as transient voltage levels and noise that may be seen by the load. The TVSS device shall be placed at interfaces between shipboard electrical power distribution systems and individually connected user equipment and distribution panels. The TVSS device shall be suitable for use on existing shipboard and land-based Navy electric power systems.

3.2 Electrical characteristics. A TVSS device should be matched by voltage, current and configuration to the electrical environment described in IEEE C62.41.1 and the operational requirements of the equipment being protected.

3.2.1 Circuit design. A TVSS device may consist of, but not be limited to, components such as transorbs, insulated gate bipolar transistors (IGBTs), metal oxide varistors (MOVs), silicon avalanche diodes (SADs), gas discharge tubes (GDTs) and passive filter components or any combination thereof as a hybrid. The TVSS device shall be one port, Class B, Type I in accordance with ASTM F1507. Lead lengths of TVSS devices shall be as short as possible in order to ensure optimum operation (see 3.2.9). The design and construction shall be in accordance with MIL-E-917.

3.2.2 Circuit configuration. The TVSS device shall be available in either single-phase or three-phase configuration, depending upon application, and is intended to be connected in parallel with the circuit being protected. The three-phase configuration shall be constructed for delta systems with three-phase wires and an equipment (chassis) ground wire, or shall be constructed for wye systems with three-phase wires, a neutral wire, and an equipment ground wire. The single-phase configuration shall be constructed for use on three-phase systems using two-phase wires and an equipment ground wire. The single-phase configuration shall be constructed for use

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on three-phase systems, properly rated for a single phase and connected across a single phase using two hot wires and an equipment ground wire ground wire.

3.2.3 Maximum continuous operating voltage (MCOV). The MCOV for the system voltages shall be as provided in Table I.

TABLE I. MCOV for line to line and line to neutral or ground.

System	MCOV Line to Line	MCOV Line to Neutral or Ground
115 Vrms	≥ 150 Vrms	≥ 150 Vrms
200 Vrms	≥ 260 Vrms	≥ 260 Vrms
440 Vrms	≥ 550 Vrms	≥ 550 Vrms

3.2.4 Minimum peak current per mode. The minimum peak current per mode of the TVSS device shall be 80,000 A. For an ungrounded delta configuration, the modes are defined as AB, AG, BC, BG, CA, and CG. For an ungrounded wye configuration, the modes are defined as AB, AN, AG, BC, BN, BG, CA, CN, CG, and NG.

3.2.5 Suppressed voltage ratings. The maximum let-through voltage shall be as shown in Table II for line-to-line (L-L) and line-to-ground (L-G) for a dynamic (power applied) TVSS device, with a 6-inch lead length when subjected to a 6 kV/3 kA Location Category B impulse waveform as defined in IEEE C62.41.2. Impulses shall be applied/measured at positive polarity at 90° voltage phase angle. Let-through voltages are measured from the positive peak of the sine wave to the positive peak of the surge. The maximum allowable UL 1449 supressed voltage ratings shall be as provided in Table II.

TABLE II. Maximum allowable UL 1449 supressed voltage ratings.

System Voltage	3 Phase Delta System (ungrounded)		3 Phase WYE System (ungrounded)		
L-L	L-L	L-G	L-L	L-N	L-G
115 Vrms	500 V	500 V	500 V	500 V	500 V
200 Vrms	900V	900 V	900 V	900 V	900 V
440 Vrms	1500 V	1500 V	1500 V	1500 V	1500 V

3.2.6 Maximum let-through voltage ratings. The maximum let-through voltage shall be as shown in Table III for line-to-line (L-L) and line-to-ground (L-G), with a 6-inch lead length when subjected to a 6 kV/3 kA Location Category B impulse waveform as defined in IEEE C62.41.2. The oscilloscope used shall have a minimum bandwidth of 100 MHz and an input impedance of 1 Mohm, capable of displaying the TVSS device entire response. Impulses shall be applied/measured at positive polarity at 90° voltage phase angle. Let-through voltages shall be measured from the positive peak of the sine wave to the positive peak of the surge. The maximum let-through voltage shall be as provided in Table III.

TABLE III. Maximum let-through voltage.

System Voltage	Let-Through Voltage	
	L-L	L-G
115 Vrms	675 V	675 V
200 Vrms	675 V	675 V
440 Vrms	1750 V	1750 V

3.2.7 Input power frequency. The TVSS device shall be fully operational within a power frequency range from 50 to 420 Hz.

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3.2.8 Response time. The TVSS device shall have a response time of less than 50 nanoseconds.

3.2.9 Protection modes. Protection shall be provided using directly-connected elements for all modes. For an ungrounded wye configuration, the TVSS device shall provide L-L, L-N, and L-G directly-connected protection elements for all phases, including N-G. For an ungrounded delta configuration, the TVSS device shall provide L-L and L-G directly-connected protection elements for all phases. This specification acknowledges TVSS using 'derived neutral' architecture, provided that L-L and L-N directly-connected protection elements are used for all phases, including N-G.

3.2.10 Circuit protection. The TVSS device shall be listed to have a short circuit current rating (SCCR) equal, or greater, than the prospective short circuit fault current available at the location where it is intended to be installed. Any "conditions of use" by which this SCCR has been listed (such as the inclusion of series fuses or circuit breakers) shall be adhered to. Each line connection from the distribution system phase conductor to the TVSS device shall be fused and properly coordinated to disconnect the TVSS device from the distribution system in the event of a TVSS failure. Individual fuses, such as VSP-type surge suppressor protection fuses, one per phase, internal to the TVSS enclosure, shall be sized to protect the feeder cable. Appropriate feeder cable such LSFSGU-9, not to exceed 18 inches in length with characteristics in accordance with 3.2.5, shall be used.

3.2.11 Status indication. The TVSS device shall provide a visual indication of power available and open fuses as described further. When power is applied, the TVSS device shall provide a power available visual indication. If a TVSS device line fuse opens, or input power is removed, the power available light shall extinguish. Status indicator voltages shall be less than or equal to 50 V. LEDs, neon lights, other lamps, etc., energized at line voltage are specifically not allowed.

3.2.12 Grounding. The TVSS device enclosure shall be connected to the ship's ground, providing an equipment ground in accordance with MIL-STD-1310.

3.2.13 Electromagnetic compatibility. The TVSS device shall comply with the EMC requirements of MIL-STD-461.

3.3 Mechanical characteristics.

3.3.1 Enclosure requirements. The TVSS device components shall be mounted in a NEMA 4 hinged enclosure with a seamless weld in accordance with NEMA 250 and the mounting shall meet the requirements of IEEE STD 45. The overall size of the enclosure shall be minimized to meet application requirements.

3.3.2 Component mounting. No components other than indicator lights shall be mounted external to the enclosure.

3.3.3 Construction. The TVSS device enclosure shall be robust in order to withstand a shipboard environment. Vibration or resilient mounts may be utilized to ensure the device components and enclosure will operate satisfactorily when tested to the shock and vibration requirements of MIL-S-901 and MIL-STD-167-1. Wire connections, terminal strips, and other device components shall not crack or become dislodged during testing. When specified, non-magnetic enclosures are allowable for minesweep ships.

3.3.4 Dimensions and weight. Due to the space restrictions aboard the ship, the overall size of the enclosure shall be minimized; and in no case shall the enclosure be larger than 30 cm (12 in) in height by 30 cm (12 in) in width by 30 cm (12 in) in depth, inclusive of all mounting hardware. The TVSS device shall weigh less than 40 pounds.

3.3.5 Painting. The TVSS device enclosure shall be painted machinery gray, in accordance with FED-STD-595/26307. The paint scheme selected by the manufacturer shall be compatible with shipboard maintenance practices as detailed in NSTM Chapter 631 and for use with Navy formula no.111 as formulated in accordance with MIL-DTL-15090. The manufacturer may select any coating scheme that matches color 26307 and is compatible with the maintenance practices in NSTM Chapter 631 and the shipboard use of Navy paint formula no. 111.

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3.3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.7 Identification plates. The TVSS device shall be identified with a label plate that shall include the following information:

a. Manufacturer's name, address, and phone number
b. Model and serial number
c. Nominal operating voltage and frequency
d. Maximum continuous operating voltage
e. Short circuit current rating
f. Maximum surge current in kA for an 8/20 microsecond waveshape
g. Intended power system configuration
h. Underwriters Laboratories listing data
i. Installation instructions
j. Warranty information

It is understood that more than one label may be utilized to provide all information specified above.

3.3.8 Workmanship. The contractor and all facilities involved in the production of the TVSS devices shall be certified as operating a quality organization in accordance with ANSI/ISO/ASQ Q9001.

3.3.9 Encapsulant. Any epoxy or encapsulant shall meet the requirements of UL 94 for reduced flammability. Additionally, the Hazardous Materials Information System (HMIS) value of the cured encapsulant shall be zero.

3.4 Environmental conditions.

3.4.1 Location. The TVSS device shall be installed in a sheltered location aboard the ship; as such, the device will be protected from the external environment. The sheltered environment may be either controlled or uncontrolled.

3.4.2 Ambient temperature. The TVSS device shall withstand storage temperatures of -25 to 70 °C (-13 to 158 °F) and ambient operating temperatures of -25 to 70 °C (-13 to 158 °F) without any derating of the device.

3.4.3 Relative humidity. The TVSS device shall operate in a relative humidity of 0 to 100 percent (non-condensing).

3.4.4 Corrosion. The TVSS device shall be corrosion-resistant in the marine environment in accordance with NEMA 250.

3.4.5 Inclination. The TVSS device shall operate at any inclination.

3.4.6 Ship motion and attitude. The TVSS device shall be fully operational for the ship motion and attitude conditions set forth in DOD-STD-1399-301A.

3.4.7 Shock. The TVSS device shall be classified as Grade A, Type 1 in accordance with MIL-S-901. It shall withstand the shock loads as specified in MIL-S-901. No portion of the device or enclosure shall break or cause a hazard to personnel.

3.4.8 Vibration. The TVSS device shall be in accordance with the Type I vibration requirements of MIL-STD-167-1.

3.5 Operation.

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3.5.1 Service life. The design service life of the TVSS device for the specified operating environment shall be a minimum of 20 years.

3.5.2 Duty cycle. The TVSS device shall be capable of continuous operation (24 hours a day, 7 days a week).

3.5.3 Status. Proper operation of the TVSS device shall be determined by self-diagnosis and indicated by a device that is externally observable (e.g.: light-emitting diode or indicator lamp).

3.5.4 Failure. In the event of internal failure, the TVSS device shall provide an external visible indication of such failure.

3.6 Reliability/maintenance.

3.6.1 Reliability. The TVSS device shall have a minimum life expectancy of 20 years.

3.6.2 Maintenance. The TVSS device shall require no preventive maintenance.

3.7 First article. When specified (see 6.2), a TVSS device shall be subjected to first article inspection in accordance with 4.2.

3.8 Performance characteristics. The TVSS device shall conform to the requirements of ASTM F1507, and shall be listed by Underwriters Laboratories, Inc. under UL 1449.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article inspection shall be performed on one of each model/configuration TVSS device when a first article sample is required (see 3.7). The inspection shall include the examination specified in 4.4 and the tests specified in 4.5. A first article inspection shall be required for vendor eligibility.

4.3 Conformance inspection. The conformance inspection shall consist of the examination specified in 4.4 and the tests specified in 4.5.

4.4 Examination. Each TVSS device shall be visually examined for compliance with the requirements of 3.3. At the contractor's option, portions of the visual examination may be performed by automated optical and electronic sensors.

4.5 Methods of inspection.

4.5.1 Electrical. The TVSS device shall be tested in accordance with and shall be fully compliant with UL 1449 performance tests. The test impulse shall be an IEEE C62.41.1 and C62.41.2 Category B 6kV/3kA Combination wave using UL 1449 paragraphs 34.4, 34.5, and Table B1.1 calibration and testing criteria. The intent of this specification is to ensure uniform testing using 100% test impulses, appropriate oscilloscope sample rates and 6" leads from outside the enclosure.

4.5.2 Life cycle. The number of applied voltage and current impulses to the TVSS device shall be 2500 each line-to-line and 2500 each line-to-ground; the impulses shall have a 1.2 x 50 microsecond waveshape and 1 kV magnitude for 115 Vrms devices and 2.5 kV for 440 Vrms devices per MIL-STD-1399-300A, applied at 90

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electrical degrees positive polarity of the steady state ac waveform. Impulses shall be applied on the order of 2 per minute. The TVSS device shall withstand life cycle testing with zero failures and less than 10 percent degradation of any parameter. Should any failures occur, the Contractor shall take corrective action and restart the life cycle test from the beginning. The TVSS device must endure the application of at least 5000 impulses (2500 line-to-line and 2500 line-to-ground) without failure.

4.5.3 Dimensions. Measure external dimensions for conformance with the requirements of 3.3.5 and verify the size of bolt holes for mounting the units. The dimensions shall be measured and compared to the dimensions specified in this specification and in the contract documents.

4.5.4 Shock. Conformance with the requirements of 3.4.7 shall be verified by testing in accordance with MIL-S-901.

4.5.5 Temperature. Temperature tests for conformance with the requirements of 3.4.2 shall be in accordance with MIL-STD-810, methods 501 and 502.

4.5.6 Vibration. For conformance with the requirements of 3.4.8, a vibration response investigation shall be performed. The TVSS device shall be subjected to endurance testing at fixed frequencies based upon the critical frequencies determined from the vibration response investigation. When the vibration response investigation does not identify critical frequencies, tests should be conducted at the maximum vibration frequency. The test duration at each frequency shall be 2 hours. The vibration frequency range and amplitude shall be as specified in Table IV.

TABLE IV. Vibration frequency range and amplitude.

Vibration Frequency Range (Hz)	Vibration Amplitude (mm)
4 to 10	0.75
11 to 15	0.75
16 to 25	0.50
26 to 33	0.25
34 to 40	0.13
41 to 50	0.08

4.6 Product verification tests. The following tests shall be performed on production units.

4.6.1 Operation. Every production TVSS device shall be powered at the factory to demonstrate that installed indicator lights and fusing are operational. Product verification testing as specified in 4.5.1 shall be accomplished at the discretion of the acquisition activity; at a minimum, every twenty-fifth production TVSS device shall undergo and pass testing.

4.6.2 Assembly. The TVSS device shall be visually inspected to ensure that all specified components are present. Where automated processes are employed for the assembly of TVSS devices, electro-optical devices may be used to perform the inspection.

4.6.3 Workmanship. Components shall be visually inspected for defects.

4.6.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified for the test method cited.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging

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requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES (This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The TVSS device specified herein is intended for the protection of sensitive shipboard electrical and electronic equipment. It is not intended to protect purely resistive loads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required (see 3.7).

c. Issue of the DoDISS to be cited in the solicitation and the specific issue of individual documents referenced (see 2.2).

d. Packaging requirements (see 5.1).

6.3 Shelf life. This specification covers items where shelf life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from *DOD 4140.27-M, Shelf-life management Manual*, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/."

6.4 Subject term (key word) listing.

Circuit protection

Electrical equipment

Electrical protection

Electronics

Custodians:
Army - MI
Navy - SH
Air Force - 11

Preparing Activity:
Navy - SH
(Project 5920-0818)

Review Activities:
Army - CR
Navy - AS, CG
Air Force - 03, 19, 99
DLA - CC

Civil Agency:
GSA - FSS
CIV - NA

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